

Redrob Matcher Engine

A Hybrid Semantic-Behavioral Candidate Discovery & Ranking System for the Founding Senior AI Engineer Role

Team ID
Antigravity-Ranker
Challenge Stack
Python + NumPy + Scikit-Learn

SLIDE 02

Traditional Recruiting Traps vs. Intelligent Discovery

⚠️ The Traditional Keyword Trap

Conventional Applicant Tracking Systems (ATS) rely on simple keyword counting, making them highly vulnerable to:

- **Keyword Stuffers:** Non-technical candidates who paste high-frequency buzzwords into descriptions to cheat search embeddings.
- **Research-Only Profiles:** Finding candidates with matching keywords (e.g., CV/Speech papers) who have zero production or IR infrastructure experience.
- **Silent Candidates:** Perfect-on-paper profiles who are entirely inactive, unavailable, or have extremely low recruiter reply rates.

💡 Our Intelligent Discovery Philosophy

We designed a multi-dimensional matching engine that aligns with the judgment of a world-class recruiter:

- **Semantic Match & Experience Curve:** Evaluates context rather than occurrences, verifying experience duration and role trajectories.

- Anti-Service Exclusions: Automatically filters candidates whose entire career is in pure consulting/service firms (TCS, Infosys, etc.).
- Behavioral Multiplier: Treats candidate availability, notice periods, and active platform engagement as key ranking modifiers.

Antigravity-Ranker

Slide 2 of 7

SLIDE 03

The Honeypot Shield: Disqualification Defense

Security Goal

Identify and block impossible candidate profiles ("honeypots") to maintain 0% trap presence in our top ranks (disqualification occurs if honeypots > 10% in top 100).

The dataset embeds synthetic trap profiles with logical contradictions. Our ranking code executes a robust pre-filter pass that checks every record for temporal and profile anomalies, forcing their scores to 0.0 immediately.

Rule 1: Startup Founding Dates

Detects candidates claiming to have started working at Indian AI startups before they were legally founded:

- Krutrim (founded April 2023)
- Sarvam AI (founded July 2023)

Result: Catches 87 impossible profiles.

Rule 2: Career Duration Mismatch

Compares declared career job durations against start and end dates.

- e.g. Start: Sept 2023, End: None (Max possible: 33 months), but declared duration is 166 months.

Result: Catches 19 duration-mismatch honeypots.

Antigravity-Ranker

Slide 3 of 7

SLIDE 04

Engine Architecture: 5-Pillar Fit Score

Pillar	Weight	Logic and Matching Mechanism
Experience Curve	15%	Non-linear fit curve. Peaks at ideal 5–9 years. Penalizes junior candidates (<4 years) and slightly discounts over-senior managers (>12 years).
Applied ML Role Fit	25%	Measures career duration in applied AI/ML titles. Scans job descriptions for key engineering/retrieval concepts. Heavily penalizes candidates in unrelated domains (marketing, HR).
Pedigree & Exclusions	20%	IT Services exclusion rule. If 100% of candidate's career is at large consulting firms (TCS, Wipro, Infosys, etc.), they receive a pedigree penalty. Rewards product-company backgrounds.
Required Skills	25%	Scans for required categories: Vector DBs (Milvus, Pinecone, FAISS), Embeddings (sentence-transformers), Python, and Eval (NDCG, MAP). Factored with endorsements & duration.
Semantic Similarity	15%	Fits a local TF-IDF model on profile summaries and work details, calculating cosine similarity against core JD concepts (dense retrieval, evaluation, search scale).

Antigravity-Ranker
Slide 4 of 7
SLIDE 05

The Behavioral Multiplier: Real Availability

Notice Period (Stated)

A candidate with 90-day notice is often too slow to hire.

- Notice $\leq$ 30 days: Multiplier = 1.0
- Notice $\leq$ 60 days: Multiplier = 0.9
- Notice $\geq$ 90 days: Multiplier = 0.35

Location & Relocate

Primary offices are in Noida and Pune.

- In Pune / Noida / NCR: Multiplier = 1.0
- Willing to Relocate: Multiplier = 0.9
- Outside India (No sponsorship): Multiplier = 0.45

Platform Engagement

Filters out candidates who logged off months ago.

- Active $\leq$ 30 days: Multiplier = 1.0
- Active $\geq$ 180 days: Multiplier = 0.3
- Response Rate $\geq$ 70%: Multiplier = 1.0

Formulation: Final Score = Base Fit Score $\times$ Notice Factor $\times$ Location Factor $\times$ Active Factor $\times$ OpenToWork Factor $\times$ GitHub Factor

Antigravity-Ranker

Slide 5 of 7

SLIDE 06

Grounded Reasoning Generator

Dynamic Fact-Based Synthesis

Our reasoning generator is fully offline and runs dynamically in milliseconds, adhering to the 6 criteria of the manual evaluation:

- No Hallucinations: Incorporates variables read directly from the candidate profile (exact companies, years, and skills).
- JD Alignment: Explicitly cites search database systems (Milvus, Pinecone) or evaluation frameworks (NDCG).
- Honest Concerns: Notes if notice periods are long, or if the candidate's career has been entirely in service firms.

- High Variation: Rotates through multiple grammatical structures to prevent robotic templates.

Sample Reasoning Output (Rank 1)

"Strong senior candidate with 6.5 years in Applied ML, highlighting hands-on expertise in Weaviate, BM25, PEFT at Razorpay. Strong platform activity signals. Short notice period (45 days) facilitates a rapid transition."

Sample Reasoning Output (Rank 3 - With Concern)

"Product-focused AI Engineer with 6.0 years of experience and deep expertise in OpenSearch, Sentence Transformers, Learning to Rank. Has shipped ranking/retrieval systems in production. However, notice period of 60 days is slightly long."

Antigravity-Ranker

Slide 6 of 7

SLIDE 07

Technical Specifications & Reproducibility

⚙️ Sandbox Performance

Designed for high performance and low operational complexity in standard Docker environments.

- Execution Time: ~74 seconds for the entire 100,000 candidate pool (Limit: 300s).
- Memory Footprint: ≤ 1.5 GB RAM (Limit: 16 GB).
- Zero Network: 100% self-contained locally; no external LLM API dependencies.
- CPU Only: Highly optimized linear vector algebra (NumPy) + Scikit-Learn TF-IDF.

✅ Submission Checklist

Verifies all challenge constraints and validator scripts pass.

- Rows Count: Exactly 100 data rows + 1 header row.
- Ranks range: Exactly 1 to 100.
- Scores: Monotonically non-increasing with rank (descending).
- Honeypot Rate: 0% trap presence in the final top-100 shortlist.
- GitHub Repository: Clean, structured codebase with strict gitignore rules.

Antigravity-Ranker

Slide 7 of 7